

Summary of "VisualWebArena: Evaluating Multimodal Agents on Realistic Visually Grounded Web Tasks"

Problem and Motivation

While most existing benchmarks evaluate text-only models, real-world computer interfaces are inherently multimodal despite the fact that humans naturally rely on both visual and textual cues during everyday tasks. To this end, this work introduces VisualWebArena, a benchmark designed to assess and advance the performance of agents that must understand both images and text while interacting with realistic, web-based environments. The motivation is twofold: (1) to simulate real human-computer interactions where visuals play an essential role, and (2) to drive the development of more robust multimodal agents capable of complex planning, reasoning, and execution on visually rich web tasks.

Solution Overview

VisualWebArena benchmark contains 910 tasks from three realistic web environments: Classifieds, Shopping, and Reddit.

1. **Realistic Web Environments:** The environments are self-hosted replicas based on real-world websites. Each environment provides a mix of text and visual content that mimics typical human interactions (e.g., product browsing, posting in forums, or searching classifieds).
2. **Task Design and Evaluation:** Tasks are created using a diverse set of intent templates and cover a range of difficulties. The evaluation framework models the environment as a partially observable Markov decision process (POMDP) and uses hand-designed reward functions. These functions assess both text outputs (via exact, must_include, and fuzzy matching) and visual tasks (using image-based metrics like SSIM for image matching).

Limitations

Complexity of Visually Dense Environments: In environments with dense visual layouts, such as the Classifieds and Reddit, the standard accessibility tree representation often fails to provide sufficient detail, which necessitates more effective representations like the proposed Set-of-Marks. However, even with SoM, significant challenges remain.

Action Planning and Error Propagation: This paper reports instances where agents successfully execute actions but then inadvertently undo them or make errors in multi-step

reasoning. This indicates that the current models are still limited in handling long-horizon action sequences and maintaining coherent plans.

Limited Generalization Across Task Difficulties: Although the benchmark includes a wide range of task difficulties, many state-of-the-art models exhibit near-zero performance on medium to hard tasks, suggesting that their reasoning and planning abilities require further improvement.

Future Research Directions

Expanding the Action Space: Future agents might benefit from more refined action representations or novel prompt strategies that better leverage the visual structure of webpages.

Robustness and Scalability: Given the substantial performance gap between current agents and human users, additional research is needed to handle diverse failure modes, extend task difficulty, and ensure reliability across different web environments.

Summary of Class Discussion

In class, we discussed how reward functions were applied in each pipeline. It was found that reward functions were designed only to the task completion aspect, which might show future directions to design fine-grained metrics for each POMDP components. Furthermore, each task description was written on a fixed template. While the task pipeline is complex enough, monotonous user queries would not simulate the realistic situation that much. Finally, we also discussed the optimal path to reach out to the task goal. Although tasks in this benchmark contain a single path, it would be another future direction to explore the most efficient paths to achieve the same goal.

Summary of “Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs”

Problem and Motivation

Most vision-language models are trained on top of language models, which means they have strong language capabilities, but their visual perception abilities are lacking. Particularly, the authors notice that visual question answering is subpar. To solve this problem, they propose two contributions: (1) a benchmark that covers the error analysis of these models, and (2) a mixup-based embeddings method.

Related Works

Previous works study other problems that vision-language models face, such as catastrophic forgetting, hallucination, vulnerability to adversarial attacks. They also posit that scale and model parameters cannot solve intrinsic issues like these. The authors also overview recent work in visual encoders, comparing and contrasting different training strategies and objective functions. However, they do not address the errors they found in the visual question answering task - things like “orientation”, “count”, or “color” of objects.

Solution Overview

As mentioned above, this paper has two contributions: the benchmark and the embedding technique. For the former, they formulate a CLIP-blind pair of data points such that the CLIP embeddings for both data points are close, but the DINOv2 embeddings are far away. For the latter, to bolster the visual perception abilities of vision language models by generating embeddings with more visual information. Particularly, they linearly interpolate vision-language embeddings from CLIP and DINOv2 to introduce stronger visual perception information into the embedding.

Limitations

The paper mostly focuses on the setting where CLIP embeddings are similar but DINOv2 embeddings are far without much explanation on why this setting was chosen. There is also not a lot of intuitive analysis of the interweaved MoF.

Future Research Directions

A potential future direction for research could involve using the benchmark for continual learning – ensuring that models are able to work for not only CLIP-blind pairs but also for CLIP-non-blind pairs. Furthermore, introducing adapter-style models that specialize in certain weaknesses of

CLIP-blind pairs can be trained and alternated between to ensure strong robustness. Finally, being able to dynamically learn the SLL ratio between CLIP and DINOv2 embeddings could yield stronger results; although, it would require more data to learn.

Summary of Class Discussion

1. This paper studies pairs with similar CLIP embeddings but different DINOv2 embeddings – what about the other way around? Shouldn't they also be part of the error analysis?
2. The paper mentions that the human annotators got the right answer to the question 95.7% of the time. Do they have an error analysis on why it is not 100%?
3. With the best results at 75% mixing, does this imply that CLIP's language abilities are stronger than its visual perception? What if we used a stronger vision model – would this percentage go down?
4. How does one intuitively interpret interleaved MoF?